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Phylogeography of the widespread creek chub *Semotilus atromaculatus* (Cypriniformes: Leuciscidae)

Susana Schönhuth¹  | Roderick B. Gagne² | Fernando Alda³ | David A. Neely⁴ | Richard L. Mayden¹ | Michael J. Blum²

¹Department of Biology, Saint Louis University, St Louis, Missouri

²Department of Ecology and Evolutionary Biology, Tulane University, New Orleans, Louisiana

³Louisiana State University, Museum of Natural Science, Baton Rouge, Louisiana

⁴Tennessee Aquarium Conservation Institute, Chattanooga, Tennessee

Correspondence

Susana Schönhuth, Department of Biology, Saint Louis University, St. Louis, Missouri.
Email: susana.schönhuth@gmail.com

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The extent and nature of genetic differentiation in *Semotilus atromaculatus*, one of the most abundant and widespread leuciscids in North America, were evaluated based on mitochondrial (mt) and nuclear DNA sequence variation. Phylogenetic relationships were first inferred based on a fragment of the cytochrome b (*cytb*) region and the nuclear intron *s7* gene for *S. atromaculatus* and all other congeners as well as representative species from all other genera in the creek chub–plagopterin clade. The phylogeography of major haplogroups of *S. atromaculatus* was also assessed according to variation in a fragment of the mitochondrial *cytb* region from 567 individuals across its range. All analyses identified *S. thoreauianus*, *S. lumbee* and *S. corporalis* as reciprocally monophyletic groups. Analyses of nuclear sequence variation resolved *S. atromaculatus* as a single clade, where *S. thoreauianus* and *S. lumbee* were recovered as the sister group to *S. atromaculatus*, and *S. corporalis* was resolved as sister to all other species in the genus. Analyses of mtDNA sequence variation recovered *S. atromaculatus* as three well supported and differentiated monophyletic groups, with a widespread genetically homogeneous lineage extending across most of the current range of the species; a more geographically restricted and geographically structured lineage in the southern Appalachians, sister group to *S. lumbee*; and a geographically restricted lineage was identified from two Gulf Slope basins. Evidence of complex mito-nuclear discordance and phylogeographic differentiation within *S. atromaculatus* illustrates that further analysis of widespread species is warranted to understand North American freshwater fish diversity and distributions.

KEYWORDS

cryptic differentiation, freshwater fish, headwater capture, mito-nuclear discordance, systematics

1 | INTRODUCTION

Leuciscids are a highly diverse family of Holarctic freshwater fishes. Though many leuciscids exhibit small ranges, sometimes limited to a single catchment or basin (e.g. *Relictus solitarius* Hubbs & Miller 1972, *Moapa coriacea* (Hubbs & Miller 1948), *Phoxinellus dalmaticus* Zupančič & Bogutskaya 2000, *Squalius torgalensis* (Coelho, Bogutskaya, Rodrigues & Collares-Pereira 1998), *Algansea avia* Barbour & Miller 1978), the family includes several species that are notably common and widespread (e.g. *Phoxinus phoxinus* (L. 1758), *Notemigonus crysoleucas* (Mitchill 1814), *Semotilus atromaculatus* (Mitchill 1818))

(Froese & Pauly, 2017). Evidence of morphological variability in widely distributed taxa, including species that have been considered monotypic based on some external characteristics (Contreras-Balderas, 1975; Johnston & Ramsey, 1990), raises the possibility, however, that widespread leuciscids are complexes of distinct lineages. Recent assessments of morphological and genetic variation that have revealed unexpected patterns of differentiation and complexity in widespread Nearctic and Palearctic groups also indicate that there is likely greater diversity in the family than is currently recognized (Blum *et al.*, 2008; Glotzbecker *et al.*, 2016; Kottelat, 2007; Palandac *et al.*, 2015, 2017; Schönhuth *et al.*, 2011, 2012, 2015). As many

widespread leuciscids are often dominant constituents of fish assemblages, this suggests that current estimates of Nearctic and Palearctic fish diversity are conservative and that there are additional evolutionary lineages than are currently recognized.

Prior studies illustrate how resolving uncertainties about widespread leuciscids can improve accounts of freshwater fish diversity and distributions. Recent assessments of morphological variability in the widely distributed Palearctic species *P. phoxinus*, for example, determined that it is composed of three previously recognized species, resurrected as *Phoxinus colchicus* Berg 1910 from the eastern Black Sea, *Phoxinus lumaireul* (Schinz 1840) from the Adriatic Sea and *Phoxinus strandjae* Drensky 1926 from the Veleka (Bulgaria) and Resowska (Turkey) drainages, and three new species (*Phoxinus bigerri* Kottelat 2007, *Phoxinus septimaniae* Kottelat 2007 and *Phoxinus strymonicus* Kottelat 2007) with restricted peri-Mediterranean distributions in Spain, France and Greece (Bogutskaya & Naseka, 2004; Kottelat, 2007). A recent molecular study of *P. phoxinus* also revealed evidence of a multispecies complex in the western Balkan Peninsula (Palandacic *et al.*, 2015). Morphological analyses of *Phoxinus* Rafinesque 1820 from Italy and the western Balkans described four additional species (Bianco & De Bonis, 2015), though some incongruencies were found in subsequent molecular analyses (Palandacic *et al.*, 2017). Studies of widely distributed Nearctic leuciscids, including species of *Camptostoma* Agassiz 1855 (Blum *et al.*, 2008; Schönhuth *et al.*, 2011), *Cyprinella* Girard 1856 (Glitzbecker *et al.*, 2016) and *Dionda* Girard 1856 (Schönhuth *et al.*, 2012), have uncovered similar evidence of unrecognized diversity, leading to the resurrection of previously described species and the discovery of new lineages that probably warrant species recognition.

Understanding the full extent and nature of leuciscid diversity requires further assessments of groups like the North American endemic genus *Semotilus* Rafinesque 1820, which has a complicated taxonomic history that hints at the presence of unrecognized diversity. The genus is currently resolved within a lineage referred to as the creek chub–plagopterin clade (recently recognized as subfamily Plagopterinae) that includes members of the genera *Couesius* Jordan 1878, *Hemitremia* Cope 1870, *Lepidomeda* Cope 1874, *Margariscus* Cockerell 1909, *Meda* Girard 1856 and *Plagopterus* Cope 1874 (Schönhuth *et al.*, 2018; Simons *et al.*, 2003; Simons & Mayden, 1997). *Margariscus* had been placed in synonymy with *Semotilus* (Hubbs & Lagler, 1958), until Coburn and Cavender (1992) recognized the validity of *Margariscus* based on osteological data. Coburn and Cavender (1992) also placed *Semotilus* as sister to *Hemitremia* and resolved *Margariscus*, *Semotilus*, *Hemitremia* and *Couesius* as paraphyletic and closely related to exoglossins rather than to plagopterins. Subsequent molecular analyses also recovered *Semotilus* as sister to *Hemitremia*, but in a clade encompassing creek chubs and plagopterins (Bufalino & Mayden, 2010; Simons *et al.*, 2003; Simons & Mayden, 1997). However, a more comprehensive phylogenetic analysis of North American leuciscids that provided further support for a creek chub–plagopterin clade recovered *Hemitremia* nested within *Semotilus* (Martin & Bonett, 2015).

The genus *Semotilus* also is an excellent subject for studying the extent and nature of leuciscid diversity because it includes geographically widespread and morphologically variable species like the creek

chub, *S. atromaculatus*, which is one of the most widely distributed species of freshwater fish in North America. Its range extends across the eastern two-thirds of the U.S.A. and southeastern Canada including the Atlantic Slope, Laurentian Great Lakes, Hudson Bay, Mississippi River and Gulf of Mexico (GOM) basins. The species ranges as far west as Saskatchewan, Canada, Wyoming and the Brazos River in Texas, U.S.A. There are also isolated populations in the upper Pecos Mountains and Canadian River systems of New Mexico, U.S.A. (Figure 1). Morphological variation reported across the range of *S. atromaculatus* includes clinal patterns in scale counts (Johnston & Ramsey, 1990) that often serve as traits for diagnosing species (Blum *et al.*, 2008). Variation in total circumference scales and predorsal scales have been detected along two north–south axes (Johnston & Ramsey, 1990), where scale counts decline from north to south along the Atlantic coast and along an interior axis running southward from the Great Lakes region. It is possible that the clines are a result of phenotypic plasticity arising from latitudinal variation in environmental conditions (Blank & Lamouroux, 2006), but the observed patterns of morphological variation might also reflect unrecognized evolutionary differentiation within the species (Johnston & Ramsey, 1990).

The complicated taxonomic history of *S. atromaculatus* offers further intimations of evolutionary differentiation within the species. The species was initially described as *Cyprinus atromaculatus* from Walkill River (Hudson River), New York, but it was later considered a junior synonym of *Semotilus corporalis* (Mitchill 1817) until a re-assessment by Bicknell and Dresslar (1885), affirmed by Jordan (1885), who adopted the specific name *atromaculatus* based on a review of specimens described as *C. corporalis*, *Cyprinus bullaris* Rafinesque 1817 and *C. atromaculatus*. Currently, the species encompasses several synonyms: *Semotilus dorsalis* Rafinesque 1820 (described from Kentucky River, Kentucky), *S. cephalus* Rafinesque 1820 (described from 'creeks of Kentucky'), *S. hammondii* Abbott 1860 ('Kansas'), *S. macrocephalus* Girard 1856 (Missouri River, South Dakota), *S. speciosus* Girard 1856 (Sweetwater River, Platte or Nebraska River), *Leucosomus pallidus* Girard 1856 (Antelope Creek, Arkansas), *L. incrassatus* Girard 1856 (Arkansas River, Oklahoma), *Leuciscus pulchelloides* Ayres 1850 (Fairfield County, Connecticut), and *L. iris* Valenciennes 1844 (described 'from New York'). Though no subspecific taxa are currently recognized, Greeley (1930) identified two subspecies based on north–south axes of clinal variation in scale counts: *S. atromaculatus atromaculatus* to the north and *S. atromaculatus thoreauianus* to the south. Greeley (1930) also suggested the existence of a third unnamed subspecies of *S. atromaculatus* from North Carolina, Georgia and Alabama, which Fowler (1945) later conflated with *S. atromaculatus thoreauianus* based on lower scale counts found in southern Atlantic Slope *S. atromaculatus*.

Patterns of morphological variation and the taxonomy of other recognized species of *Semotilus* similarly allude to the presence of greater evolutionary diversity within the genus. Besides *S. atromaculatus*, the genus includes *Semotilus thoreauianus* Jordan 1877 (Dixie chub), *Semotilus lumbee* Snelson & Suttkus 1978 (Sandhills chub) and *S. corporalis* (fallfish). *Semotilus thoreauianus*, which was described from the Flint River (Apalachicola River), Georgia, was considered a subspecies of *S. atromaculatus* or a clinal variant (Bailey *et al.*, 1954; Jordan & Evermann, 1896; Williams, 1965) until its re-description by Johnston and

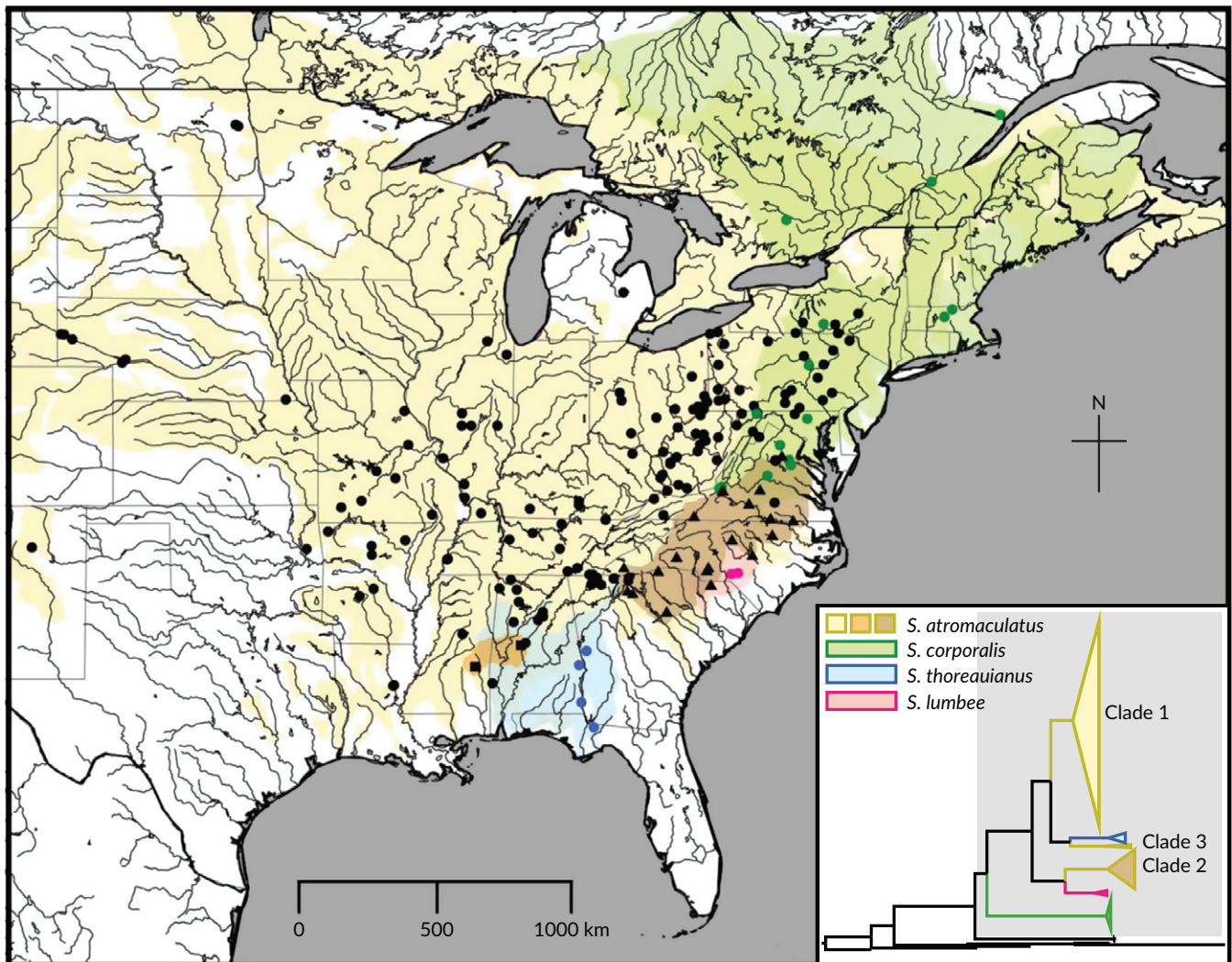


FIGURE 1 The location of sampling sites (●, ▲, ■) across the eastern U.S.A. (Supporting information listed in Table S1). The brown, orange and yellow-shaded regions collectively represent the distributional boundary of *Semotilus atromaculatus* according to the currently accepted taxonomy of the genus. The yellow-shaded region (enclosing black circles) depicts a widespread mitochondrial clade identified within *S. atromaculatus*, whereas the orange and brown-shaded regions (enclosing black triangles and squares) represent two divergent mitochondrial clades identified within *S. atromaculatus*. The blue-shaded region represents the range of *S. thoreauianus*, the pink-shaded region represents the range of *S. lumbee*, and the green-shaded region represents the range of *S. corporalis*. The inset box shows the phylogenetic relationships among major clades inferred from the mitochondrial *cytb* gene

Ramsey (1990). It inhabits drainages across the GOM Slope from the Ochlockonee River system in Georgia and Florida to the Tombigbee River system in Alabama (Figure 1). It is morphologically distinguishable and putatively reproductively isolated from *S. atromaculatus* in the Black Warrior and Cahaba River basins though morphological evidence of introgression has been detected in some areas of co-occurrence (Johnston & Ramsey, 1990). *Semotilus atromaculatus* exhibits an atypical snout and dorsal fin position as well as other attributes similar to *S. thoreauianus* in the upper Tallapoosa River (Johnston & Ramsey, 1990). Johnston and Ramsey (1990) also detected morphological variation within *S. thoreauianus*, suggesting that the population in the Ochlockonee drainage is divergent from populations elsewhere. *Semotilus lumbee*, which was described from Aberdeen Creek (Pee Dee River) in North Carolina, is endemic to the Sandhills region of North Carolina and South Carolina. The range of *S. lumbee* encompasses portions of

the Cape Fear, Pee Dee and Santee River drainages (Figure 1). Based on tuberculation, habitat and distribution, Snelson and Suttkus (1978) proposed that *S. lumbee* is sister to *S. atromaculatus*. *Semotilus corporalis*, which has been consistently distinguished from *S. atromaculatus*, is endemic to the north-eastern U.S.A. and eastern Canada including Hudson Bay, Lake Ontario and Atlantic Slope drainages from Quebec, New Brunswick and Ontario (Canada) to the James River drainage in Virginia (Figure 1). The species was first described as *Cyprinus corporalis* Rafinesque 1817 from Walkill River (Hudson River), New York. Four species have since been synonymized with *S. corporalis*: *Cyprinus bullaris* Rafinesque 1817 (Hudson River, New York), *Leuciscus storeri* Valenciennes 1844 (New York), *Leuciscus argenteus* Fitzinger 1832 (James River, Virginia) and *Leucosomus rhotheus* Cope 1861 (Trenton, New Jersey).

The aim of this study was to assess the nature of genetic and phylogenetic differentiation within and among all currently recognized

members of the genus *Semotilus*. The primary objective of the study was to determine whether *S. atromaculatus* is monophyletic or a more complex aggregation of evolutionary lineages by examining range-wide patterns of genetic variation and by reconstructing the phylogeny of the genus using complementary molecular approaches. Doing so not only addressed longstanding questions about the taxonomy and relationships among members of the genus, it also illustrated how comprehensive analysis of ostensibly widespread species can improve estimates of freshwater fish diversity and distributions in North America.

2 | MATERIAL AND METHODS

2.1 | Fish collection and DNA sequencing

Tissue was obtained from 523 individuals of *S. atromaculatus* sampled across 148 localities within its range (Figure 1 and Supporting Information Table S1). Tissue was also obtained from another 43 individuals of the other three congeneric species sampled from 22 localities. Thus, 567 individuals from 163 streams were sampled for this study (Figure 1 and Supporting Information Table S1). Either dorsal or caudal-fin tissues were sampled, after which fish were returned to their place of capture. All tissues were then preserved in 95% ethanol. When whole bodies were taken, fish were euthanized through institutionally approved use of MS-222 and either preserved whole in 95% ethanol, frozen whole upon capture, or fixed in 10% formalin after tissue samples were taken and later transferred to 95% ethanol (Saint Louis University Institutional Animal Care and Use Committee protocols 1,550, 1,626, 1,595, 2,135, 2,134, 2,467; EPA QAPP 6497/1/1; Lazorchak *et al.*, 1998). When whole bodies were preserved, voucher individuals were deposited in ichthyological collections at Saint Louis University, St. Louis, Missouri (SLUM, STL), University of Alabama Ichthyological Collection, Tuscaloosa, Alabama (UAIC), North Carolina State Museum of Natural Sciences, Raleigh, North Carolina (NCSM), and Royal Ontario Museum, Toronto, Canada (ROM).

Genomic DNA was extracted from all tissue samples using commercial extraction kits (DNeasy Blood & Tissue Kit; Qiagen; www.qiagen.com). The mitochondrial cytochrome *b* (*cytb*) region and the unlinked nuclear *S7* ribosomal protein gene (*s7*) were sequenced and analyzed, as both regions have proven to be informative in prior phylogenetic and phylogeography studies of other North American leuciscid species (Schönhuth *et al.*, 2011, 2016). Mitochondrial (mt)DNA and nuclear (n)DNA sequence variation was characterized for the same specimens. Analyses of the mtDNA *cytb* region focused on a fragment of 929 base pairs (bp) for 199 selected individuals across the distribution of *S. atromaculatus* and other members of the genus (hereafter referred to as the long-fragment dataset) and a fragment of 486 bp for an additional 336 *S. atromaculatus* and nine *S. corporalis* compiled with 199 trimmed long-fragment sequences, for a total of 567 individuals from 163 localities (hereafter referred to as the short-fragment dataset). Primers GLU 5'-TGACTTGAAGAAC-CACCGTT-3' and THR 5'-ATCTTCGGATTACAAGACCG-3' were used to generate sequences for both *cytb* datasets (Bagley *et al.*, 2004). Amplifications that failed to show product were repeated

with the degenerate GLU primer GLUDG 5'-TGACTTGAARAAC-CAYCGTTG-3' (Bagley *et al.*, 2004). A fragment of 811 bp of the *s7* gene that included the first intron was sequenced as described by Chow and Hazama (1998). Because nuclear genetic variation among species is typically lower than mitochondrial genetic variation (Schönhuth *et al.*, 2014, 2016), we selectively obtained *s7* sequence data for a subset of 99 specimens from 59 localities representing all lineages derived from the topology revealed by the aforementioned *cytb* analyses. When amplification was weak or when double bands were recovered, a nested PCR was performed for the *s7* region with two internal primers as described in Schönhuth *et al.* (2011, 2012). All PCR reactions were carried out using a Peltier Thermal Cycler-200 (MJ Research; www.first-in-technology.com). Primers for direct sequencing of the purified PCR product were the same as those used for PCR amplification. Sequencing of purified PCR products was carried out at Retrogen Inc. (www.retrogen.com), at the U.S. Environmental Protection Agency (www.epa.gov; Bagley *et al.*, 2004) and Tulane University (New Orleans, Louisiana). Representative sequences obtained for this study have been deposited in GenBank under accession numbers MH053592–MH054258 (Supporting Information Table S1).

The resulting DNA sequences were aligned and analyzed as three separate datasets: a short-fragment *cytb* dataset (567 specimens, 486 bp), a long-fragment *cytb* dataset (199 specimens, 929 bp) and an *s7* dataset (99 specimens, 811 bp). No gaps were found in the *cytb* datasets, whereas the *s7* alignment included several indels with lengths ranging from 1 to 21 bp. All positions were included in analyses of sequence variation. Phylogenetic relationships were inferred for each dataset using maximum likelihood (ML) and Bayesian inference (BI) approaches. Phylogenetic trees using ML were estimated using RAxML 7.0.4 (Stamatakis, 2006). Searches for the optimal ML trees and bootstrap support were performed using the GTR + I + G model with a mixed model of nucleotide substitution with four discrete rate categories. Inferences included partitions by codon position for the *cytb* region, while *s7* was treated as a single partition. Akaike information criterion (AIC) as implemented in MODELTEST 3.4 (Posada & Crandall, 1998) was used to identify the optimal evolutionary model for each partition of the *cytb* datasets. Robustness of the inferred trees was evaluated using bootstrap analysis based on 1,000 pseudoreplications conducted with RAxML 7.0.4 (Felsenstein, 1985; Stamatakis *et al.*, 2008). Resulting trees were imported into PAUP*4.0.b10 (Swofford, 2001) to obtain the consensus tree. BI analyses were conducted using Mr. Bayes 3.1.2 (Huelsenbeck & Ronquist, 2001). For each BI analysis, 6 million generations were implemented in four simultaneous Monte-Carlo Markov chains, with sampling of the Markov-chain set at intervals of 100 generations. Log-likelihood stability was attained after 100,000 generations. The first 10,000 trees were discarded as burn-in in each analysis. The remaining trees were used to compute a 50% majority rule consensus tree in PAUP*. Support for BI tree nodes was determined based on values of Bayesian posterior probabilities. For brevity, only ML trees are presented here, though posterior probability (PP) values and ML bootstrap support (BS) have been provided on well-supported [PP > 90 / BS > 75%] nodes in each tree. Phylogenetic analyses also included archival *cytb* or *s7*

TABLE 1 Comparison of genetic variation in major lineages in *Semotilus* spp. based on sequence variation across the mitochondrial *cytb* region

	n, n_{Loc}	n_{Dr}, B	S	h, H_D	k	p_i (%)	PP/BS	p (%)
Lineages: <i>cytb</i> (929 bp; 30.9%; 28.2%)								
<i>S. atromaculatus</i> (all)	164, 79	33, 13	240	100, 0.989	37.61	4.054	— ^a	0.00–11.94
<i>S. atromaculatus</i> clade 1	136, 65	35, 7	138	82, 0.986	13.16	1.417	100/78	0.00–3.55
<i>S. atromaculatus</i> clade 2	25, 12	10, 8	108	17, 0.953	32.96	3.549	100/100 ^b	0.00–5.38
<i>S. atromaculatus</i> clade 3	3, 2	2, 2	36	2, 0.667	24.00	2.583	100/98 ^c	0.00–3.87
<i>S. thoreauianus</i>	7, 4	4, 2	28	5, 0.905	13.42	1.445	100/100	0.00–2.61
<i>S. corporalis</i>	24, 12	9, 4	14	12, 0.946	3.92	0.422	100/100	0.00–0.96
<i>S. lumbee</i>	4, 2	1, 1	26	3, 0.222	13.16	1.417	100/99	0.00–2.79
Total	199, 99	40, 19	287	121, 0.992	60.26	6.487	100/100 ^d	0.00–15.60
Lineages: <i>cytb</i> (486 bp; 36.8%; 32%)								
<i>S. atromaculatus</i> (all)	523, 148	43, 17	158	165, 0.951	13.92	2.864	—/— ^a	
<i>S. atromaculatus</i> clade 1	468, 130	32, 11	110	133, 0.939	5.62	1.15	—/—	
<i>S. atromaculatus</i> clade 2	52, 19	15, 10	79	30, 0.940	16.43	3.38	75/74 ^e	
<i>S. atromaculatus</i> clade 3	3, 2	2, 2	20	2, 0.667	13.33	1.293	100/92 ^f	
<i>S. thoreauianus</i>	7, 4	4, 2	16	3, 0.762	8.09	1.66	78/97	
<i>S. corporalis</i>	33, 15	9, 4	8	6, 0.763	1.78	0.36	74/100	
<i>S. lumbee</i>	4, 2	1, 1	14	2, 0.500	7.00	1.44	75/100	
Total	567, 163	53, 23	179	176, 0.957	19.95	4.107	73/87 ^d	

Note. The Lineage row gives the number of base pairs, 929 v. 486 bp, in the alignment, followed by per cent variable sites and per cent parsimony-informative sites. n , Number of individuals sequenced; n_{Loc} , number of localities; n_{Dr} , number of major rivers drainages; B , number of basins; S , number segregating sites; h , number haplotypes; H_D , haplotype diversity; k , average number of nucleotide differences; p_i , % nucleotide diversity; range of divergence based on p , % uncorrected- p distances are indicated for each lineage.

^a *S. atromaculatus* not resolved, included *S. thoreauianus* and *S. lumbee*.

^b Clade 2, sister to *S. lumbee* (PP100/BS98%).

^c Clade 3, sister to *S. thoreauianus* (PP68/BS52%).

^d *Semotilus* clade, including *Hemitemia flammea* clade.

^e Clade 2 sister to *S. lumbee* (PP78/BS98%).

^f Clade 3 sister to *S. thoreauianus* (PP77/BS75%).

data (Genbank accessions AY281053-54, JX443011, JX443045, JX443062 and GU134215-16, GU134195, GU134229, JX443236, JX443279, JX443285 respectively; Supporting Information Table S1) from representative species in all of the other six genera in the creek chub–plagopterin clade (Bufalino & Mayden, 2010; Martin & Bonett, 2015; Simons *et al.*, 2003).

Standard indices of genetic variability such as the number of segregating sites (S), haplotype diversity (H_D), nucleotide diversity (p_i , %) and average number of nucleotide differences (k), were calculated for each major lineage and for all specimens of *Semotilus* using DnaSP 5.10.1 (Librado & Rojas, 2009). Estimates of observed genetic divergence provided herein are uncorrected p -distances based on the large fragment of the *cytb* gene. Haplotype number and haplotype frequencies within the genus *Semotilus* were calculated for both *cytb* datasets. Unrooted networks for both mtDNA datasets were constructed to illustrate relationships between haplotypes. Networks were constructed using NETWORK 4.6.1.1 (www.fluxus-engineering.com) (Bandelt *et al.*, 1999) with the median-joining algorithm using maximum parsimony post-processing, where minimum-spanning trees are combined in a single network and median vectors (mv), which represent missing intermediate haplotypes, and are added to the network based on a parsimony criterion. We used a default weight of 10 for all characters and epsilon was set to 0 in order to minimize alternative median networks.

3 | RESULTS

3.1 | Mitochondrial sequence and haplotype variation

Relatively high estimates of genetic variation were recovered across the mtDNA *cytb* region (Table 1). The percentage of variable sites (36.8 and 30.9%) and parsimony informative characters (32 and 28.2%) were comparably higher for the short-fragment dataset that included a larger number of samples (Table 1). High haplotype diversity (H_D) and nucleotide diversity (p_i) were recovered for both mtDNA datasets (0.957 v. 0.992 and 6.478 v. 4.107, respectively). Among all members of the genus, *S. atromaculatus* exhibited the largest number of haplotypes, followed by *S. corporalis* (Table 1). The highest p_i , S and k values also were observed for *S. atromaculatus*, while the lowest values for these parameters were observed for *S. corporalis* in both *cytb* datasets (Table 1). Estimates of genetic divergence based on the long-fragment dataset ranged from 0.00 to 15.60% among members of the genus (Table 1). The greatest range of genetic divergence values was observed between specimens of *S. atromaculatus* (0.00–11.94%). Mean values of genetic divergence (Table 2) between recognized species ranged from 7.90–11.90% (*S. atromaculatus* and *S. thoreauianus*) to 12.90–14.40% (*S. atromaculatus* and *S. corporalis*).

Similar haplotype networks were recovered for both *cytb* datasets. As expected, the short-fragment dataset contained less

TABLE 2 Mean uncorrected *p*-distances (in percentages $\pm$ s.d.) among and within (in bold) major clades of *Semotilus*

	<i>S. atromaculatus</i> clade 1 (n = 136)	<i>S. atromaculatus</i> clade 2 (n = 25)	<i>S. atromaculatus</i> clade 3 (n = 3)	<i>S. thoreauianus</i> (n = 7)	<i>S. lumbee</i> (n = 4)	<i>S. corporalis</i> (n = 24)	<i>H. flammea</i> (n = 4)
<i>S. atromaculatus</i> clade 1	1.40 $\pm$ 0.20						
<i>S. atromaculatus</i> clade 2	10.60 $\pm$ 0.80	3.60 $\pm$ 0.40					
<i>S. atromaculatus</i> clade 3	8.40 $\pm$ 0.80	10.30 $\pm$ 0.90	2.60 $\pm$ 0.40				
<i>S. thoreauianus</i>	7.90 $\pm$ 0.70	11.90 $\pm$ 0.90	8.10 $\pm$ 0.80	1.40 $\pm$ 0.30			
<i>S. lumbee</i>	10.10 $\pm$ 0.90	9.20 $\pm$ 0.80	11.40 $\pm$ 1.00	11.70 $\pm$ 1.00	1.40 $\pm$ 0.30		
<i>S. corporalis</i>	13.20 $\pm$ 1.00	14.40 $\pm$ 1.10	12.90 $\pm$ 1.10	13.20 $\pm$ 1.10	13.70 $\pm$ 1.10	0.40 $\pm$ 0.10	
<i>Hemitremia flammea</i>	13.10 $\pm$ 1.10	14.60 $\pm$ 1.00	12.60 $\pm$ 1.00	13.90 $\pm$ 1.10	15.10 $\pm$ 1.20	14.10 $\pm$ 1.10	0.10 $\pm$ 0.10

Note. s.d. calculated according to 500 bootstrap replicates for the long fragment mtDNA *cytb* dataset.

information and exhibited lower levels of differentiation within and among haplogroups (Figure 2(a), (b)). Both networks resolved *S. thoreauianus* and *S. lumbee* haplogroups as more proximate than the *S. corporalis* haplogroup relative to *S. atromaculatus* haplogroups. Both networks also recovered three differentiated haplogroups of *S. atromaculatus*: a large and homogeneous haplogroup, which included 468 and 136 specimens based on the short and long fragment respectively, hereafter referred to as clade 1; a highly heterogeneous haplogroup, which included 52 and 25 specimens respectively, hereafter referred to as clade 2; a small haplogroup of three specimens in both datasets, hereafter referred to as clade 3. Notably, the haplotype network based on the short-fragment dataset resolved the two haplotypes observed for clade 3 almost equally distant to each other than to clade 1, whereas the haplotype network based on the long-fragment dataset resolved the haplotypes equidistant to each other and *S. thoreauianus*.

Though clade 1 in *S. atromaculatus* exhibited the highest haplotype diversity among the recovered haplogroups, it was relatively homogeneous, with most haplotypes separated by only one or a few mutations. A value of $H_D = 0.939$ was recovered for 468 specimens from 130 localities and a value of $H_D = 0.986$ was recovered for 136 specimens from 65 localities in the two respective *cytb* datasets. The haplogroup included specimens primarily from rivers in the Mississippi Basin including the Tennessee, Ohio, Cumberland, Missouri, White, Arkansas, upper and lower Mississippi rivers. The haplogroup also included specimens from the Chesapeake Bay basin, the Albemarle Sound basin, the Mobile Bay basin, the Atchafalaya Basin and the Savannah Basin, as well as rivers draining to Hudson Bay, Delaware Bay, Lake Huron and Lake Erie (Figures 1 and 2(a), (b)). Instances of more extensive genetic differentiation within clade 1 largely reflect outlier specimens from areas in the Tennessee, Cumberland, Coosa, Ouachita, or Tombigbee rivers (Figure 2(a), (b)).

Individuals in clade 2 were recovered from rivers across the southern Appalachian mountain range and adjoining Atlantic Slope in North Carolina, South Carolina, Virginia, West Virginia and Georgia (Figures 1 and 2(a), (b)). This haplogroup was the most heterogeneous with values of $H_D = 0.940$ and $H_D = 0.953$ as well as values of $p_1 = 3.38$ and $p_1 = 3.549$ for 52 and 25 specimens in the two respective *cytb* datasets. It also exhibited more extensive phylogeographic structure than the other haplogroups. Within this haplogroup, specimens from Atlantic Slope and Gulf Coast drainages, including the Santee,

Pee Dee, Cape Fear, Savannah, Pamlico, Apalachicola, Roanoke and James rivers, exhibited haplotypes that were recovered in reciprocally monophyletic sub-haplogroups. Similar haplotypes were observed between specimens collected from the upper Ohio River and the Pee Dee River and between specimens from the upper Tennessee Basin and the Santee River. A few distinctive individuals were sampled from the upper Tar River in the Pamlico Sound basin and the Meherrin River in the Albemarle Sound basin.

Haplogroup clade 3 included three *S. atromaculatus* individuals sampled from two GOM Slope localities in the Cahaba (Alabama) and Pascagoula (Mississippi) rivers.

The haplogroups composed of *S. thoreauianus* and the haplogroup composed of *S. corporalis* were both resolved closer to clade 1 in the network based on the short-fragment dataset and equidistant between clade 1 and clade 3 in the network based on the long-fragment dataset (Figure 2(a), (b)). The haplogroup composed of *S. lumbee* was proximate to clade 2 in both networks (Figure 2(a), (b)).

3.2 | Phylogenetic analysis of mitochondrial sequence variation

All phylogenetic analyses based on mtDNA sequence variation failed to recover *S. atromaculatus* as a monophyletic group. Phylogenetic analyses based on the long-fragment dataset resolved *S. atromaculatus* as a well-supported clade that nested within reciprocally monophyletic *S. lumbee* and *S. thoreauianus* clades (Figure 3). Consistent with the recovered haplotype networks, phylogenetic analyses based on the long-fragment dataset resolved *S. atromaculatus* as three well supported clades: clade 1 (PP100/BS78%) that included 136 specimens from 65 localities; clade 2 (PP100/BS100%) that included 25 specimens from 12 localities in the southern Appalachian Mountains, which was the sister group (PP100/BS98%) to *S. lumbee*; and clade 3 (PP100/BS100%) that included three specimens from two localities in the Cahaba and Pascagoula rivers, which was recovered with weak support (PP68/BS52%) as sister to the *S. thoreauianus* clade (PP100/BS100%) (Figure 3). While *S. corporalis* was resolved as the most divergent lineage in the genus (Figure 3), we did not find strong support for a sister relationship between *S. corporalis* and all the other species of *Semotilus*. *Semotilus corporalis* was resolved in a trichotomy with *Hemitremia* and a clade formed by all the other species of *Semotilus* when

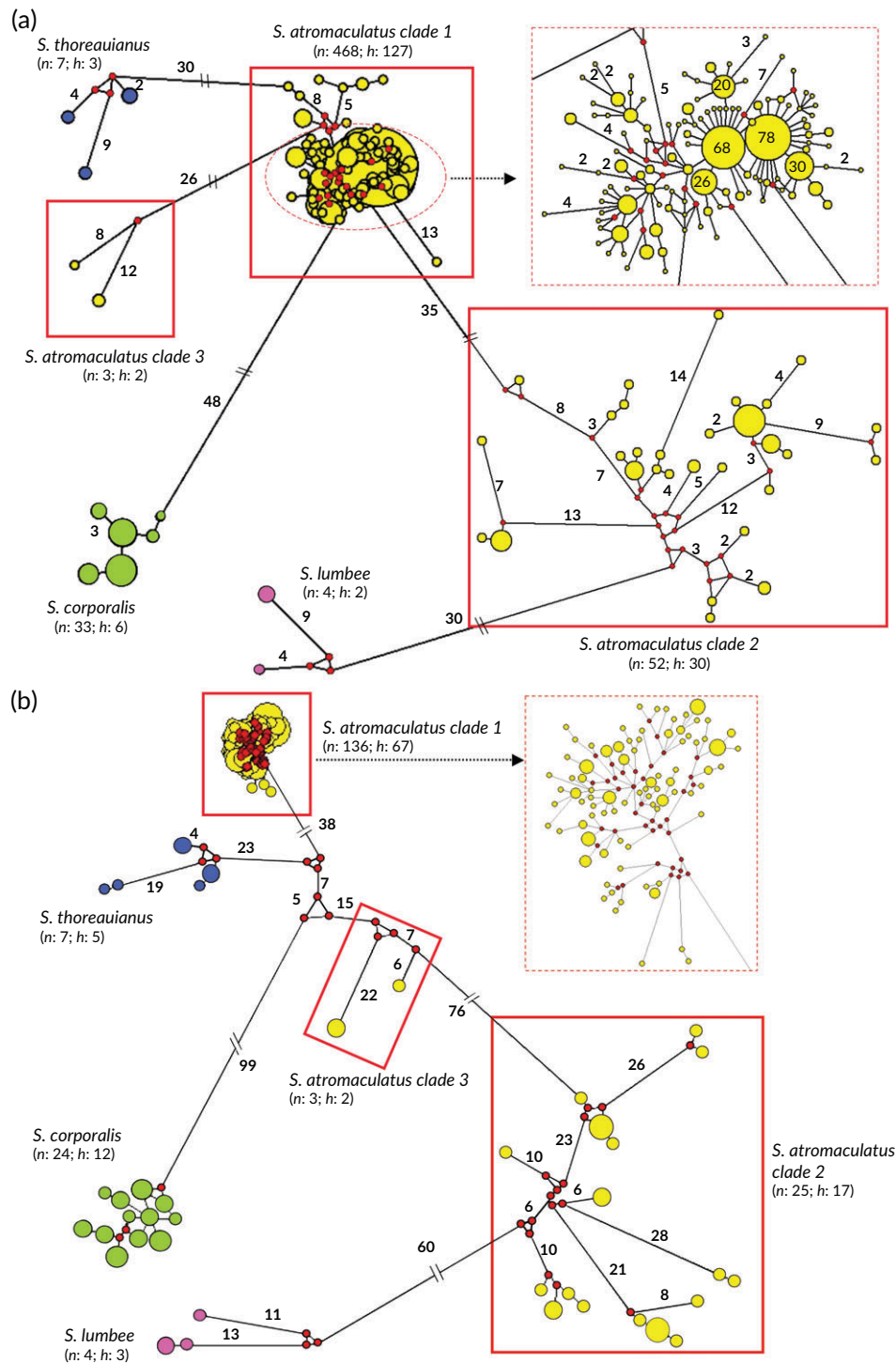


FIGURE 2 Median-joining haplotype networks for *Semotilus* spp. based on (a) a short (486 bp) and (b) a long (929 bp) fragment of the mtDNA *cytb* region. Each circle represents a different haplotype; circle sizes are proportional to the number of individuals exhibiting a particular haplotype. Recognized species are represented by different colors. ●, Missing (i.e. unsampled or extinct) haplotypes. Lines between circles represent one mutational step, and numbers presented are the number of mutations between haplotypes. Red boxes enclose three major lineages resolved in phylogenetic analyses. The red dotted-squares depict clade 1 in detail (insets)

four other genera of the creek chub–plagopterin clade were included in the analyses (Figure 3).

The phylogenetic analyses based on the short-fragment dataset were mostly congruent with those of the long-fragment dataset. One notable exception is that only two differentiated clades were resolved within *S. atromaculatus*. The support for a separate and

distinct clade 1 was only recovered in analyses of the long-fragment dataset. Despite much larger sample sizes, phylogenetic analyses (particularly BI analyses) based on the short-fragment dataset recovered weaker support for most of the clades. For example, lower support was recovered for the hierarchical structure of the four clades nesting within a wider clade that included all other 468 specimens of

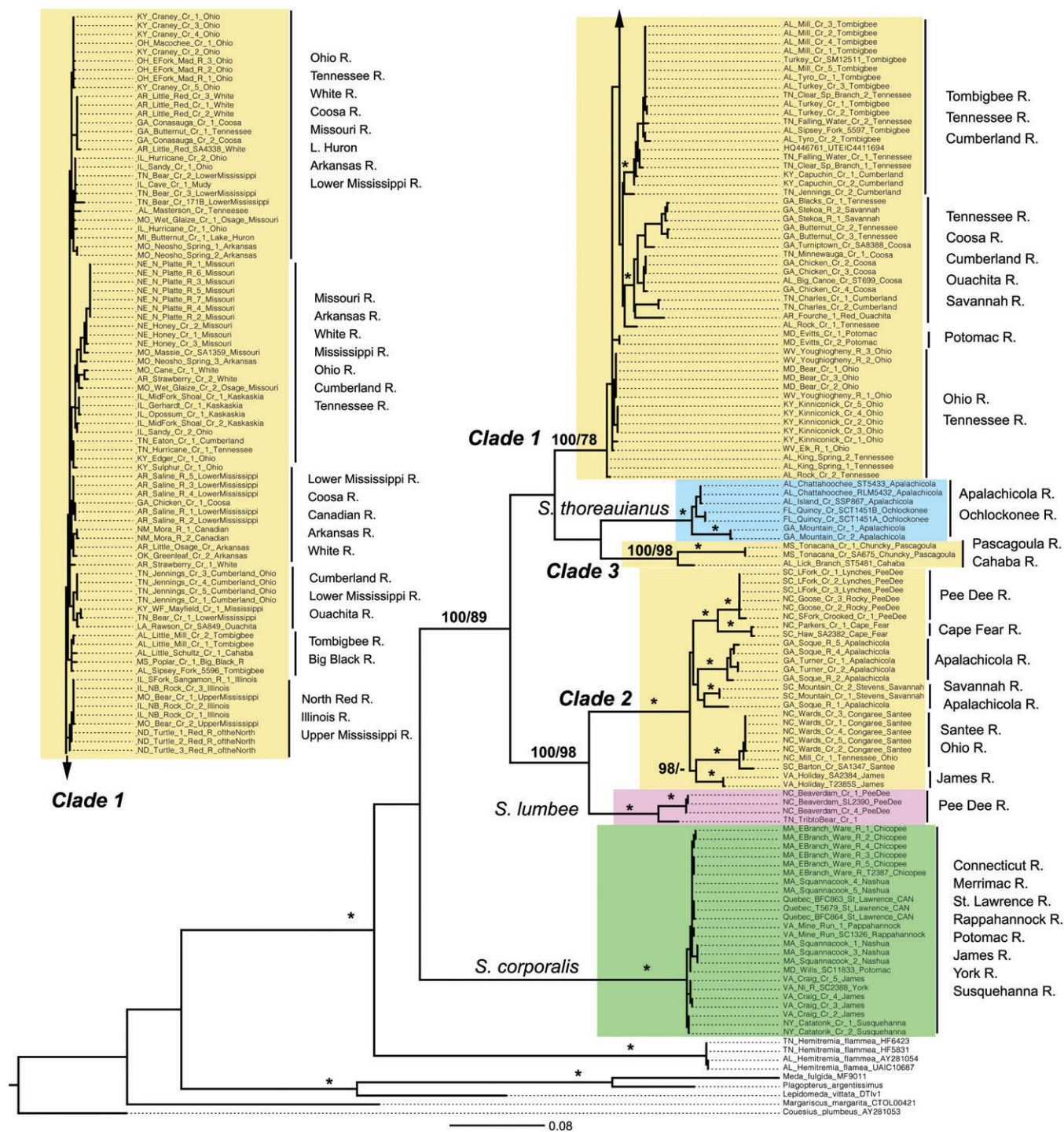


FIGURE 3 Phylogenetic relationships among 199 selected individual *Semotilus atromaculatus* sampled from 99 localities across its distribution (Figure 1 and Supporting Information Table S1), including all congeners, and the six other genera in the creek chub–plagopterin clade, based on sequence variation across 929 bp of the mitochondrial *cytb* region. A maximum likelihood (ML) tree is presented, with branch lengths measured according to the number of substitutions per site. Support for nodes is presented according to Bayesian inference followed by ML bootstrap values. *Nodes with bootstrap support (BS) and posterior probability (PP) > 95%. L, Lake; R, river

S. atromaculatus from 130 localities (PP74/BS64%). In analyses of the short-fragment dataset, clade 1 was nested in a group that also contained clade 2, clade 3, *S. thoreauianus* and *S. lumbee*. Clade 2 (PP75/BS74%) was recovered sister (PP75/BS98%) to *S. lumbee* (PP75/BS100%), while *S. thoreauianus* (PP78/BS95%) was sister (PP77/BS75%) to clade 3 (PP100/BS92%). Clade 2, which was

composed of 52 specimens from 19 localities sampled from rivers along the southern Appalachian Mountains, was largely structured according to drainage. However, as in the haplotype network, Clade 2 specimens from the upper Ohio and Pee Dee rivers were closely related, and specimens from the upper Tennessee River were nested within the Santee River lineage.

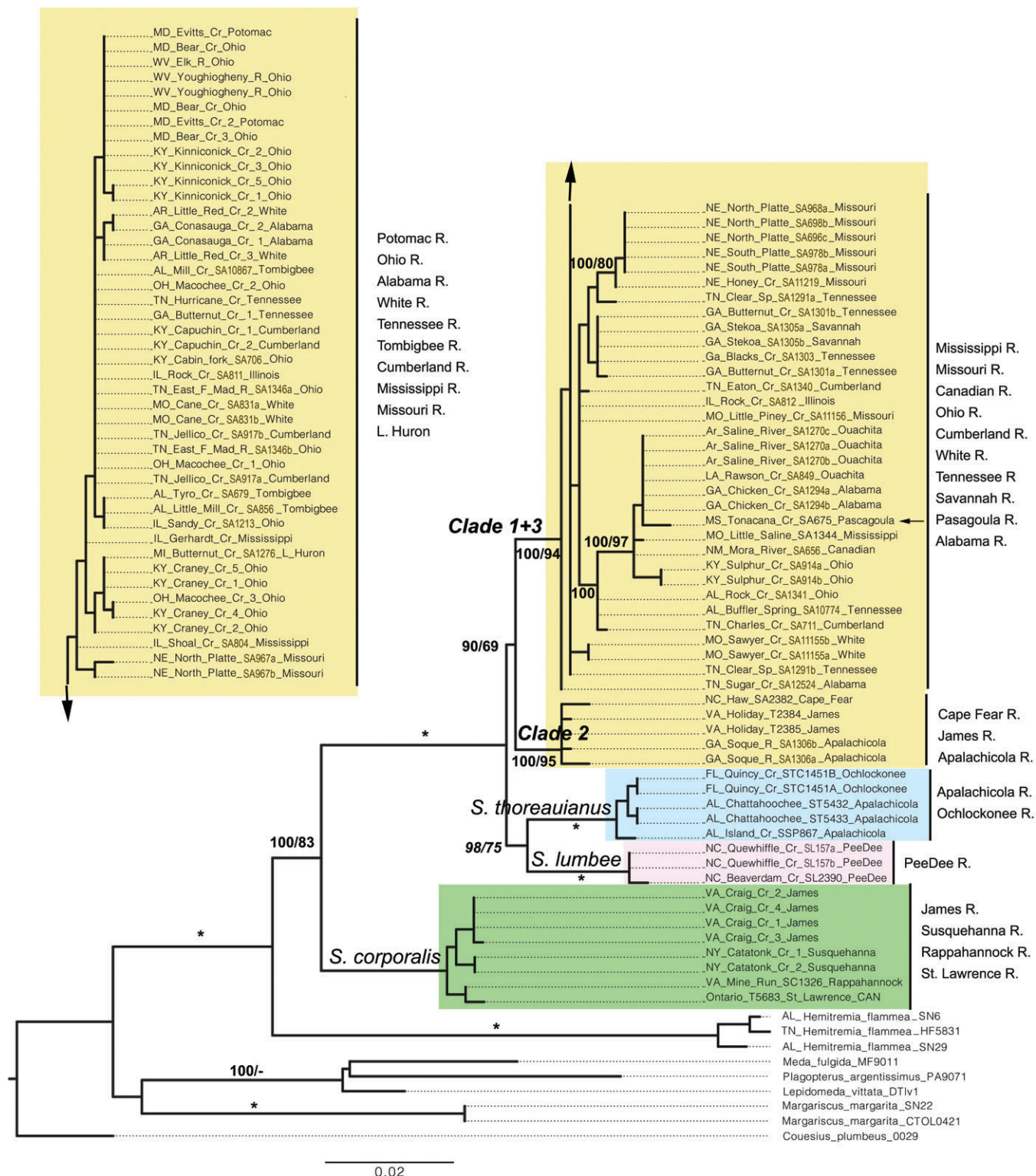


FIGURE 4 Phylogenetic relationships among 99 selected individual *Semotilus atromaculatus* sampled from 57 localities across its distribution (Figure 1 and supporting information, Table S1), including all congeners and the six other genera in the creek chub-plagopterin clade, based on sequence variation across 811 bp of the nuclear *s7* gene. A maximum likelihood (ML) tree is presented, with branch lengths measured according to the number of substitutions per site. Support for nodes is presented according to Bayesian inference followed by ML bootstrap values*. Nodes with bootstrap support (BS) and posterior probability (PP) > 95% are indicated. Cr, Creek; L, lake; R, river

3.3 | Phylogenetic analysis of nuclear sequence variation

Phylogenetic analyses of sequence variation across the nuclear *s7* region resolved *Semotilus* as a well-supported and monophyletic genus

(PP100/BS83%) sister to *Hemitremia* (PP100/BS100%) (Figure 4). None of the indels observed in *s7* were species specific within *Semotilus*. Contrary to analyses of mtDNA sequence variation, analyses of nDNA sequence variation resolved all four recognized species of

Semotilus as reciprocally monophyletic groups, though relatively weak support (PP90/BS69%) was found for the *S. atromaculatus* clade. The *S. atromaculatus* clade was resolved as sister to a clade that included a well-supported *S. thoreauianus* clade (PP98/BS75%) sister to a well-supported *S. lumbee* clade. Analyses of the *s7* region also resolved *S. corporalis* as the earliest diverging lineage sister to all the other members of the genus (Figure 4). Within *S. atromaculatus*, clade 1 (PP100/BS94%) and clade 2 (PP100/BS95%) were resolved as sister groups (PP90/BS69%). The only specimen analyzed from the mtDNA clade 3 was deeply nested within clade 1.

4 | DISCUSSION

Prior studies of morphological variation and the taxonomic history of *Semotilus atromaculatus* both allude to the possibility of undiagnosed evolutionary diversity within the genus. As has been found in other ostensibly widespread species of Nearctic freshwater fishes (Blum *et al.*, 2008; Schönhuth *et al.*, 2012), we found evidence indicating *S. atromaculatus* is a complex of evolutionary lineages that exhibit distinct geographies and distinct patterns of within-lineage genetic variation. Not only does this shed new light on the taxonomy and systematics of *Semotilus*, it also illustrates that additional analysis of widespread species like *S. atromaculatus* could improve understanding of freshwater fish diversity and distributions in North America.

4.1 | Genetic and phylogenetic variation

Observed levels of genetic variation differ among *Semotilus* lineages. Despite being widely distributed, *S. corporalis* exhibits comparably low levels of genetic diversity and differentiation. It is possible that the observed levels of homogeneity are a consequence of post-glacial range expansion across North America (Berendzen *et al.*, 2008; Schönhuth *et al.*, 2016). Testing this hypothesis would require more thorough comparison of *S. corporalis* inhabiting unglaciated and glaciated localities. Similarly, little variation was detected in *S. lumbee* and *S. thoreauianus*. This might be a consequence of the relatively restricted distributions of both species, but it might also be due to the small number of samples that were examined for this study. Resolving this uncertainty will require more thorough assessment of genetic variation in both species. Notably, higher levels of genetic diversity were detected in *S. atromaculatus* than in all other species in the genus, though the extent of genetic variation in *S. atromaculatus* differed by lineage and geography.

Phylogenetic analyses of mitochondrial and nuclear sequence variation resolved specimens of *S. lumbee*, *S. thoreauianus* and *S. corporalis* in reciprocally monophyletic groups (Figures 3 and 4). The recovered phylogenetic hypotheses affirm prior inferences that *S. thoreauianus* is distinct from *S. atromaculatus*, though no evidence was found that *S. thoreauianus* from the Ochlockonee River are more strongly differentiated from other *S. thoreauianus* as indicated by patterns of morphological variation (Johnston & Ramsey, 1990). *Semotilus corporalis* was consistently recovered as the earliest diverging clade, which parallels cladistic relationships inferred from nest building behavior and barbels (Johnston & Ramsey, 1990). Analyses of the

nuclear *s7* gene also yielded strong support for *S. corporalis* being sister to other species of the genus, as well as for monophyly of *Semotilus* and a sister relationship to *Hemitremia*. Support for this, however, was not found in analyses of mtDNA sequence variation, probably because the faster mutation rate of the mitochondrial *cytb* region impeded resolution of deeper nodes in the phylogeny due to saturation and accumulation of homoplasious substitutions.

4.2 | Evidence of mito-nuclear discordance

Phylogenetic analyses of mitochondrial and nuclear sequence variation offered mixed support for the monophyly of *S. atromaculatus*. Analyses of mtDNA sequence variation did not recover *S. atromaculatus* as a monophyletic group. Rather, specimens of *S. atromaculatus* were resolved in three well-differentiated mtDNA haplogroups that corresponded to three well-supported but non-reciprocally monophyletic mtDNA lineages. Nodal support and haplotype diversity were lower in analyses of shorter fragments, despite much larger sample sizes and higher percentages of variable sites and parsimony informative sites. Analyses of nuclear sequence variation, on the other hand, recovered *S. atromaculatus* as a monophyletic group (Figures 3 and 4). Additionally, only two major lineages were resolved within *S. atromaculatus*, though this may have been an artefact of only one individual from the third mtDNA lineage being included in the analyses.

Evidence of mito-nuclear discordance in *S. atromaculatus* could be the result of mitochondrial introgression (Ballard & Whitlock, 2004; Chan & Levin, 2005; Rhymer & Simberloff, 1996). Mitochondrial introgression can lead to complete genome replacement or mitochondrial capture, where the mitochondrial genome of a species is replaced by the mitochondrial genome of another species. Mitochondrial introgression is more likely than the reverse, in part because mtDNA is haploid and maternally inherited and thus has an effective population size 25% the size of the nuclear genome. Accordingly, it is expected to exhibit faster coalescence (Ballard & Whitlock, 2004; Chan & Levin, 2005). Though we cannot exclude other possibilities, such as a member of the genus being a paraspecies that has given rise to geographically peripheral daughter lineages, mitochondrial introgression is a more parsimonious explanation than other processes that can result in mito-nuclear discordance, especially considering the extent to which it has been detected in other freshwater fishes (Freyhof *et al.*, 2005; Nevado *et al.*, 2009; Tang *et al.*, 2012) including other cyprinoids, which are particularly prone to hybridization (Aboim *et al.*, 2010; Costedoat *et al.*, 2007; Geiger *et al.*, 2016; Perea *et al.*, 2016; Sinama *et al.*, 2013). Evidence of mito-nuclear discordance is also consistent with prior reports of hybridization among *Semotilus* species. For example, the recovered sister relationship between mtDNA clade 3 of *S. atromaculatus* and *S. thoreauianus*, as well as the absence of a distinct clade 3 in the recovered nuclear phylogeny, parallel morphological evidence of hybridization between *S. atromaculatus* and *S. thoreauianus* (Johnston & Ramsey, 1990). This suggests that further work is warranted to assess whether mito-nuclear discordance in *Semotilus* has arisen due to hybridization. Excellent first steps would be further study of *Semotilus* in and approximate to potential hybrid zones like the upper Tallapoosa River in the Alabama River basin,

where *S. atromaculatus* are morphologically similar to *S. thoreauianus* (Johnston & Ramsey, 1990).

4.3 | Comparison of molecular and morphological variation

Observed patterns of morphological variation loosely correspond to the geographic distributions of lineages within *S. atromaculatus*. Johnston and Ramsey (1990) identified two morphological clines in *S. atromaculatus*, one along the Atlantic coast and an interior cline from the Great Lakes southward that is bisected by the Appalachian Mountains. The cline along the Atlantic coast could reflect transitional sampling across the respective distributions of clade 1 and clade 2. A clear geographic separation was observed between clade 1 and clade 2 in the southern Appalachians, largely differentiating populations in Mississippi River tributaries versus Atlantic Slope rivers that drain opposite sides of the mountain range. However, the separation appears to fade northward of the Pamlico Sound Bay basin. Co-occurrence within basins also possibly underlie the observed cline in morphological variation. Though three large river systems harbor both clade 1 and clade 2, the clades do not occur in the same region in each respective drainage (Figure 1). Clade 1 occurs in the upper region of the Savannah River basin in the Sketoa River, whereas other areas of the basin harbour clade 2. In contrast, clade 1 is more widely distributed in both the Ohio and Tennessee river basins, where clade 2 exhibits restricted distributions in the upper New River in the Ohio and the Cullajasa River in the Tennessee. However, individuals from both clades were found to co-occur at three Atlantic Slope locations, in the upper Tar River (Pamlico Basin) and the Meherrin River (Chowan Basin) in North Carolina, as well as the upper Rappahannock River in Virginia, which possibly reflects secondary contact following allopatric divergence. The interior cline might similarly reflect transitional sampling across the distributions of clade 1 and clade 3, though it is also possible that it corresponds to geographic variation within clade 1. Further analysis will be necessary to determine how morphological variation relates to the distribution of *S. atromaculatus* lineages, as observed instances of separation and overlap might in part reflect sampling artefacts (the northerly reduction in geographic separation of clade 1 and clade 2 could be due to progressively denser sampling from south to north).

It is also possible that instances of headwater capture contribute to the observed patterns of variation in *S. atromaculatus* morphology. We recovered close relationships between specimens in both clade 1 and clade 2 in geographically proximate rivers that drain in opposing directions. For example, specimens of clade 1 from the Potomac River, which drains to the Atlantic Ocean, resolved among specimens from the upper region of the Ohio River, which drains to the Mississippi River. Similarly, individuals of clade 2 sampled from the headwaters of the Tennessee River, a tributary of the Mississippi River, in North Carolina nested among individuals sampled from the Santee River, which drains to the Atlantic Ocean. Samples from headwaters of the Ohio River in Virginia also nested among specimens from the Pee Dee River, which drains to the Atlantic Ocean. These findings are consistent with geological evidence of river capture in the southern Appalachian Mountains (Gallen, 2018), as well as cross-divide relationships in

the salamander genus *Eurycea* (Kozak *et al.*, 2006) and other freshwater fishes in the region, including darters (*e.g.* *Etheostoma nigrum* Rafinesque 1820), sculpins (*e.g.* *Cottus bairdii* Girard 1850) (Kinziger & Raesly, 2001; Stauffer *et al.*, 2016) and leuciscids (*e.g.* *Pimephales notatus* Rafinesque 1820; Schönhuth *et al.*, 2016), which collectively support the hypothesis of Pleistocene headwater captures (Hocutt *et al.*, 1978, 1986; Kim & Conway, 2014).

4.4 | Diversity within the genus *Semotilus*

Despite evidence of mito-nuclear discordance, the phylogeographic structure of *S. atromaculatus* illustrates that there is greater evolutionary diversity within *Semotilus* than has been previously recognized. Consideration should be given to recognizing *S. atromaculatus* as an aggregation of at least two, if not three highly differentiated evolutionary lineages exhibiting mean divergence estimates of 8.4–10.6%, which are comparable to estimates of divergence between recognized species within the genus (Table 2) as well as estimates of divergence between recognized species in other Nearctic leuciscid genera such as *Dionda* Girard 1856 (Schönhuth *et al.*, 2012), *Camptostoma* Agassiz 1855 (Blum *et al.*, 2008; Schönhuth *et al.*, 2011), *Codoma* Girard 1856 (Schönhuth *et al.*, 2015) and *Pimephales* Rafinesque 1820 (Schönhuth *et al.*, 2016). The inferred phylogeny and divergence estimates indicate that clade 1 warrants recognition as a widely distributed, genetically homogeneous lineage that spans most of the established range of *S. atromaculatus*. The majority of *S. atromaculatus* specimens included in our study were resolved within clade 1, including individuals from independent and geographically distant rivers, ranging from the Red River of the north, south to the Tombigbee River and from the Pecos River in the far western U.S.A. to the Roanoke River in the east (Figure 1). Similarly, our findings offer support for recognizing clade 2 as a geographically restricted (Figure 1), genetically heterogeneous lineage that harbours distinct sub-lineages limited to drainages across the southern Appalachian Mountains and adjoining regions of the Atlantic Slope including the Pee Dee, Cape Fear, Savannah, Apalachicola, Santee, Rappahannock-Roanoke-James, Pamlico and Chowan rivers. Analyses of mtDNA sequence variation suggest the presence of another geographically restricted lineage (clade 3) that appears to be limited to GOM Slope drainages, including the Mobile River basin and the nearby Pascagoula River drainage (Figure 1). More extensive sampling of GOM Slope drainages and further analyses of nuclear sequence variation will be necessary, however, to confirm the presence and distribution of this lineage. It is also possible that additional work might unearth evidence indicating that lineages within *S. atromaculatus* warrant recognition as distinct species (Blum *et al.*, 2008; Glotzbecker *et al.*, 2016; Schönhuth *et al.*, 2011).

Evidence of unrecognized diversity within the genus *Semotilus* underscores how examining widespread taxa can improve understanding of North American freshwater fish diversity and distributions. Not only do our findings suggest that current diversity estimates are conservative, they also help direct attention to regions that probably harbour additional unrecognized diversity. Consistent with patterns observed in other widespread groups that have been subject to phylogeographic analyses, such as *Camptostoma* (Blum *et al.*, 2008), our results affirm that the southern Appalachian Mountains

probably host additional narrowly endemic evolutionary lineages that warrant species recognition. Our findings also further highlight how idiosyncracies of contemporary distributions can reflect localized geological processes, like headwater capture across North American divides. This illustrates the value of drawing comparisons between the distribution of *Semotilus* lineages and those of other fish and stream-dependent biota (Kinziger & Raesly, 2001; Kozak et al., 2006; Schönhuth et al., 2016; Stauffer et al., 2016), which could potentially reveal the magnitude and timing of capture events. Similarly, drawing comparisons between *Semotilus* and other widespread groups could offer additional perspective on continental scale phenomena, such as the tempo, pace and magnitude post-glacial recolonization, that have shaped the historical biogeography of North American fishes (Berendzen et al., 2003, 2008; Blum et al., 2008; Near et al., 2001; Schönhuth et al., 2016).

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Author's contributions

SS: ideas, data generation, data analyses and manuscript preparation.

RG: data generation and manuscript editing.

FA: map drawing and manuscript editing.

DAN: specimen collection and manuscript editing.

RLM: manuscript editing and funding.

MJB: ideas, data generation, manuscript preparation and funding.

ORCID

Susana Schönhuth  <https://orcid.org/0000-0002-6202-4943>

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SUPPORTING INFORMATION

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